DATE	28-02-2026
TEAM ID	LTVIP2026TMID****
PROJECT NAME	HEMATOVISION: ADVANCED BLOOD CELL CLASSIFICATION USING TRANSFER LEARNING
MAXIMUM MARKS	4 MARKS

3.4 TECHNOLOGY STACK

HARDWARE AND SOFTWARE REQUIREMENTS

For the successful development, training, and deployment of HematoVision: Advanced Blood Cell Classification Using Transfer Learning, appropriate hardware and software resources are required. These requirements ensure efficient processing, accurate model training, and smooth system performance.

1. Hardware Requirements:

- **Processor (CPU)::**
  A minimum of Intel Core i5 (8th generation or above) or equivalent AMD processor is required. A higher-end processor such as Intel Core i7/i9 improves model training speed and multitasking performance.

- **RAM:**
  At least 8 GB RAM is required for handling image datasets and running deep learning frameworks. However, 16 GB RAM or more is recommended for smoother training and faster data processing.

- **Storage:**
  A minimum of 256 GB storage is required to store datasets, trained models, libraries, and project files. SSD storage is highly recommended over HDD for faster read/write speed and improved performance.

- **Graphics Processing Unit (GPU):**
  An NVIDIA GPU with CUDA support (such as GTX/RTX series) is recommended for accelerating deep learning model training. GPU significantly reduces training time compared to CPU-only processing. Although optional, GPU support enhances efficiency, especially when working with large datasets.

- **Display and Input Devices:**
  A standard high-resolution monitor is required to clearly visualize microscopic blood cell images. Keyboard and mouse are necessary for development and system interaction.

2. Software Requirements:

- **Operating System:**
  The system can run on Windows 10/11, Linux (Ubuntu recommended), or macOS. Linux is often preferred for deep learning environments due to better GPU compatibility.

- **Programming Language:**
  Python (Version 3.8 or above) is used as the primary programming language due to its simplicity, extensive libraries, and strong support for machine learning and deep learning.

- **Deep Learning Frameworks:**
  TensorFlow and Keras are used to build and implement transfer learning models. These frameworks provide pre-trained CNN architectures and efficient model training capabilities.

- **Supporting Libraries:**

- NumPy – For numerical computations.
- Pandas – For dataset handling and data manipulation.
- Matplotlib / Seaborn – For visualization of results and performance metrics.
- OpenCV – For image preprocessing and enhancement.
- Scikit-learn – For evaluation metrics and data splitting.

- **Development Environment (IDE):**
  Jupyter Notebook, PyCharm, or Visual Studio Code (VS Code) can be used for coding, debugging, and experimentation.

- **Optional Web Deployment Tools:**
  If the system is deployed as a web application, Flask or Django can be used to create a user-friendly interface for uploading images and displaying predictions.